

# Trust & Safety Moderation Action Prediction using Logistic Regression

## Project Overview

This project develops a Logistic Regression model to predict moderation actions using behavioral signals commonly observed in Trust & Safety operations.

Unlike purely rule-based systems, this dataset introduces uncertainty and overlapping behavioral patterns to better simulate real-world moderation environments.

Predicted moderation actions include:

- Allow
- Warn
- Restrict
- Ban

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## Dataset Description

The dataset contains 2,000 moderation records designed to approximate realistic Trust & Safety workflows.

### Features

| Feature             | Description                                            |
|---------------------|--------------------------------------------------------|
| User ID             | Unique user identifier                                 |
| Violative           | Indicates whether the current activity violates policy |
| Frequency           | Number of violations committed                         |
| Timestamp (minutes) | Time between incidents                                 |
| Previous Violations | Historical violation count                             |
| Severity            | Severity level of the incident                         |
| Action              | Moderation action taken (Target Variable)              |

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## Problem Statement

Predict the appropriate moderation action based on behavioral and historical user signals.

This problem is formulated as a multiclass classification task.

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## Data Preprocessing

The following preprocessing steps were performed:

- Removed `User ID`
  - Encoded categorical variables
  - Scaled numerical features using `StandardScaler`
  - Applied stratified train-test split
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## Logistic Regression Model

```
model = LogisticRegression(max_iter=20000)
```

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## Model Evaluation

The model was evaluated using:

- Accuracy
- Confusion Matrix
- Precision
- Recall
- F1-score
- Classification Report

### Performance

| Metric   | Value      |
|----------|------------|
| Accuracy | ~67% - 75% |

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## Key Findings

- Moderation decisions are influenced by multiple behavioral signals.
  - No single feature completely determines the enforcement action.
  - The dataset contains ambiguity and overlapping classes, resulting in more realistic performance.
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## Dataset Characteristics

### Action Distribution

| Action   | Approximate Percentage |
|----------|------------------------|
| Allow    | 43%                    |
| Warn     | 23%                    |
| Restrict | 18%                    |
| Ban      | 16%                    |

This distribution reflects typical platform moderation scenarios where most users remain compliant.

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### Strengths

- More realistic than rule-based datasets.
  - Includes behavioral ambiguity.
  - Produces believable model performance.
  - Suitable for demonstrating model evaluation and error analysis.
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### Limitations

The dataset is still synthetic and does not include:

- Actual user-generated text
  - Human disagreement
  - Contextual understanding
  - Sarcasm
  - Appeals and policy exceptions
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### Technologies Used

- Python
  - Pandas
  - NumPy
  - Scikit-learn
  - Matplotlib
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## Future Improvements

- Incorporate NLP-based content signals.
  - Evaluate on human-annotated moderation datasets.
  - Compare multiple machine learning algorithms.
  - Build a moderation dashboard using Streamlit.
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## Conclusion

This project demonstrates the application of Logistic Regression in Trust & Safety moderation and provides a more realistic approximation of human moderation behavior by introducing uncertainty and overlapping decision boundaries.